03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 18

(JCD 12.3)

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Previous lecture

- Core He fusion happens in a convective region and somewhat repeats the process of H core burning
- If core He is exhausted, there is an inert C-O core and 2 shell burning regions
- C-O Core becomes degenerate for stars less massive than 10 solar masses
- Nuclear fusion ends and the star moves up the AGB
- Through not-well understood processes, the star loses its mass outside the C-O core, revealing a planetary nebula and finally a white dwarf

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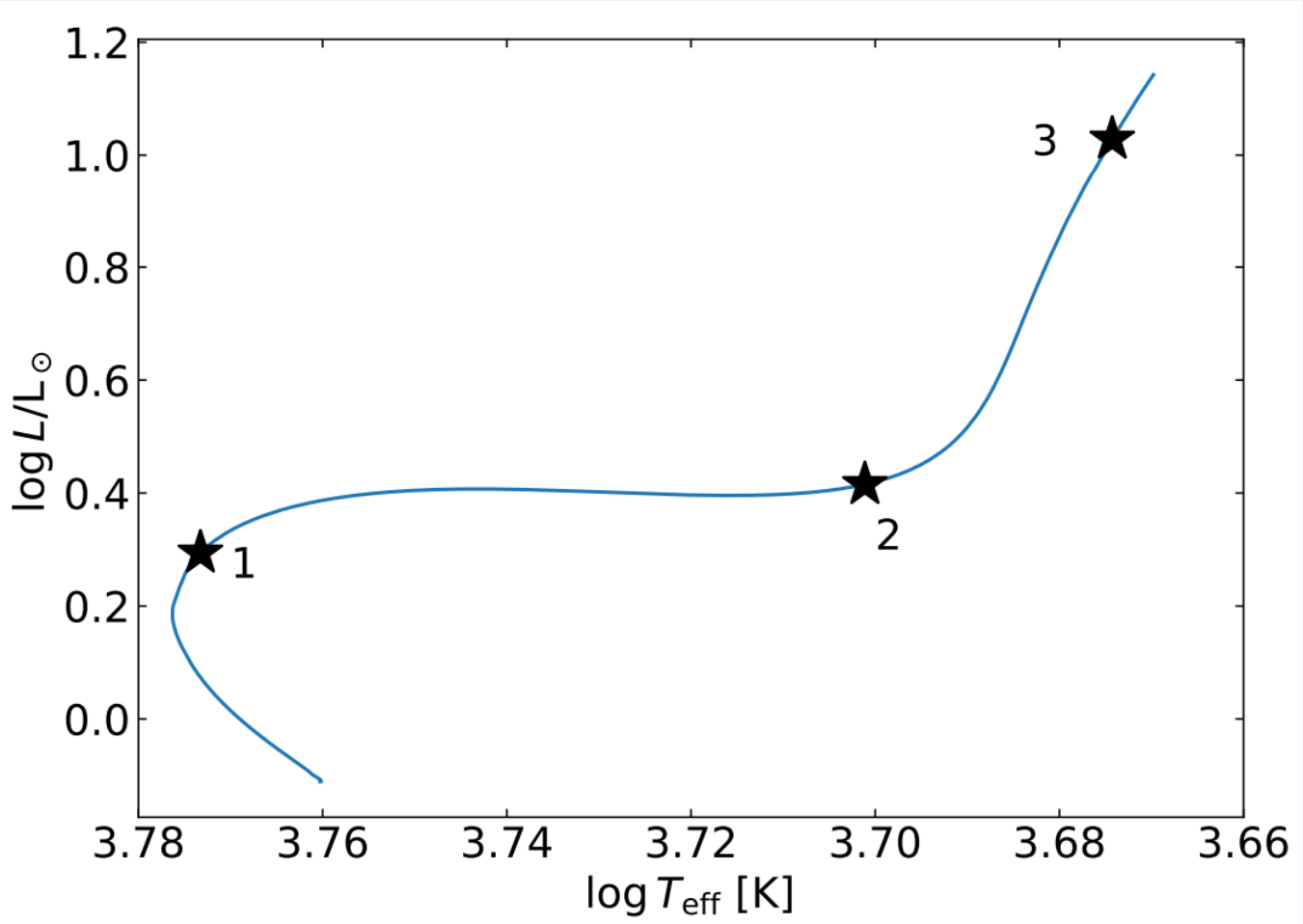
Outline

Low-mass star evolution
- End of core H burning
- Shell H burning
- Degenerate core
- He flash
- Evolution after core He burning

Summary

Low-mass star evolution: MS to RGB

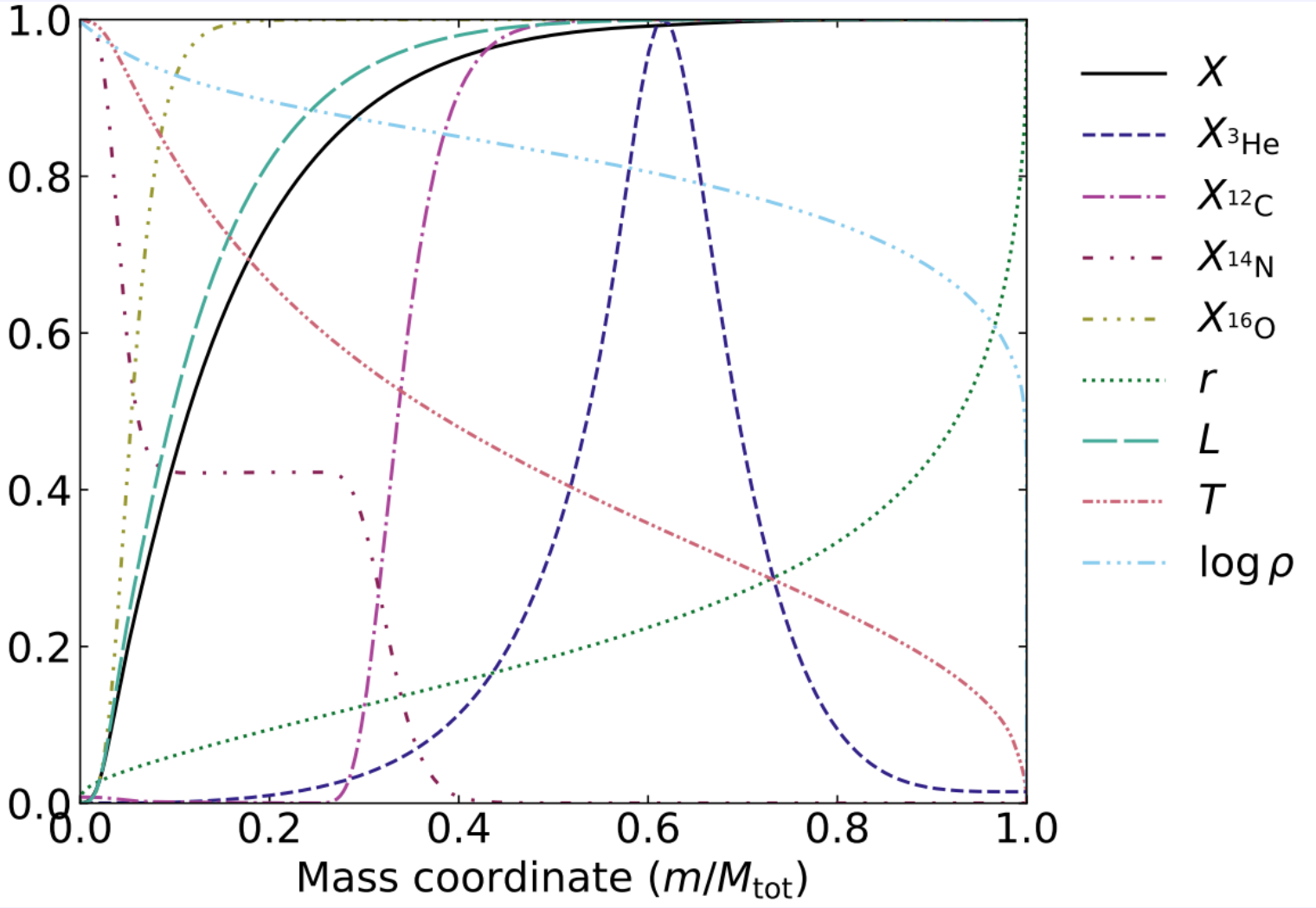


Credit: JCD - Fig 12.11

Core H fusion - difference with stellar mass

- Remember lecture 10 & 15: low mass stars have no convective core (or thus mixing) during the MS
- H used up gradually in the core
- So the transition from core to shell H burning also goes gradually
- Originally, the H shell is thick, as it extends almost to the centre

End of core H fusion (point 1)

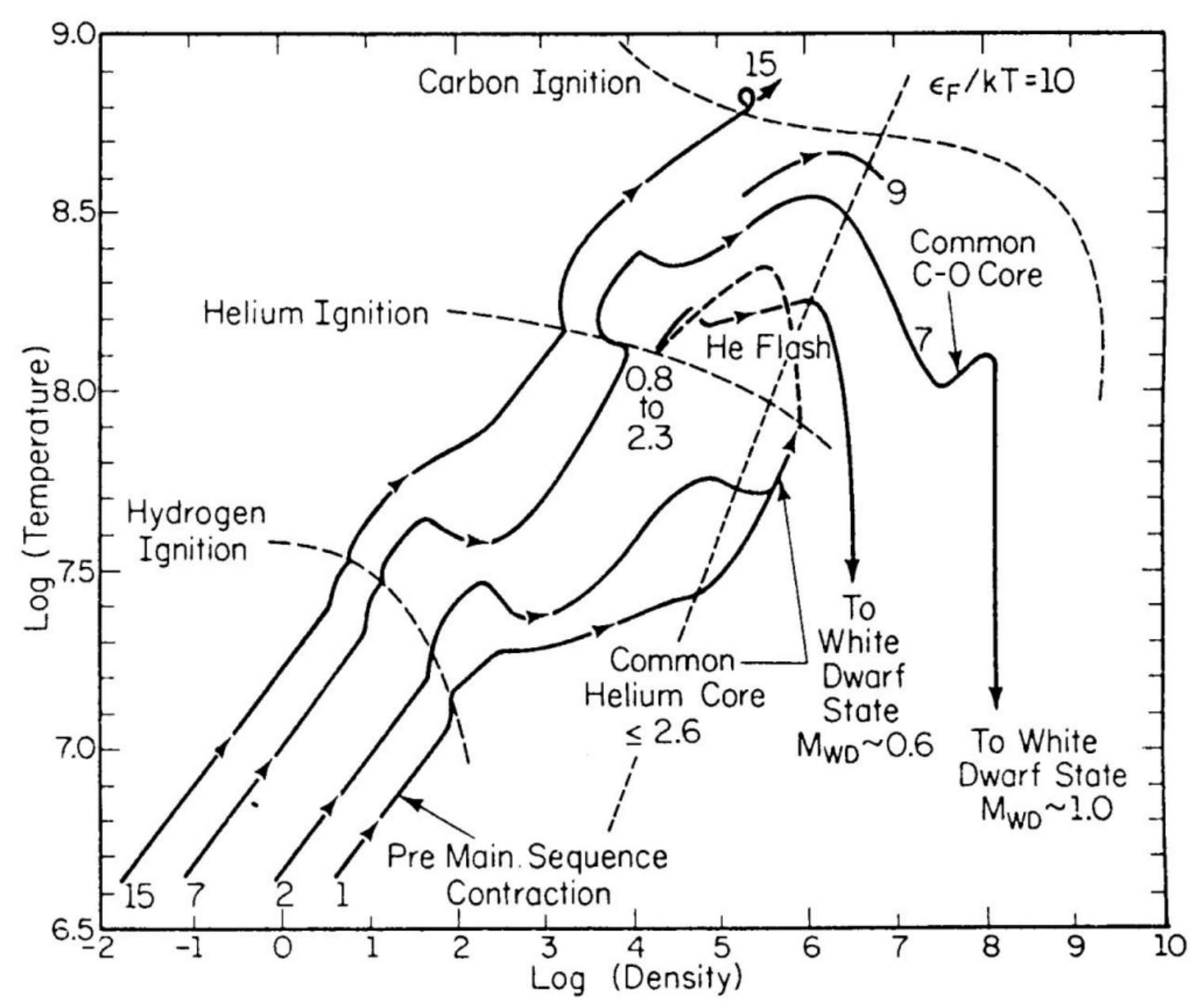


Credit: JCD - Fig 12.12

Shell H fusion

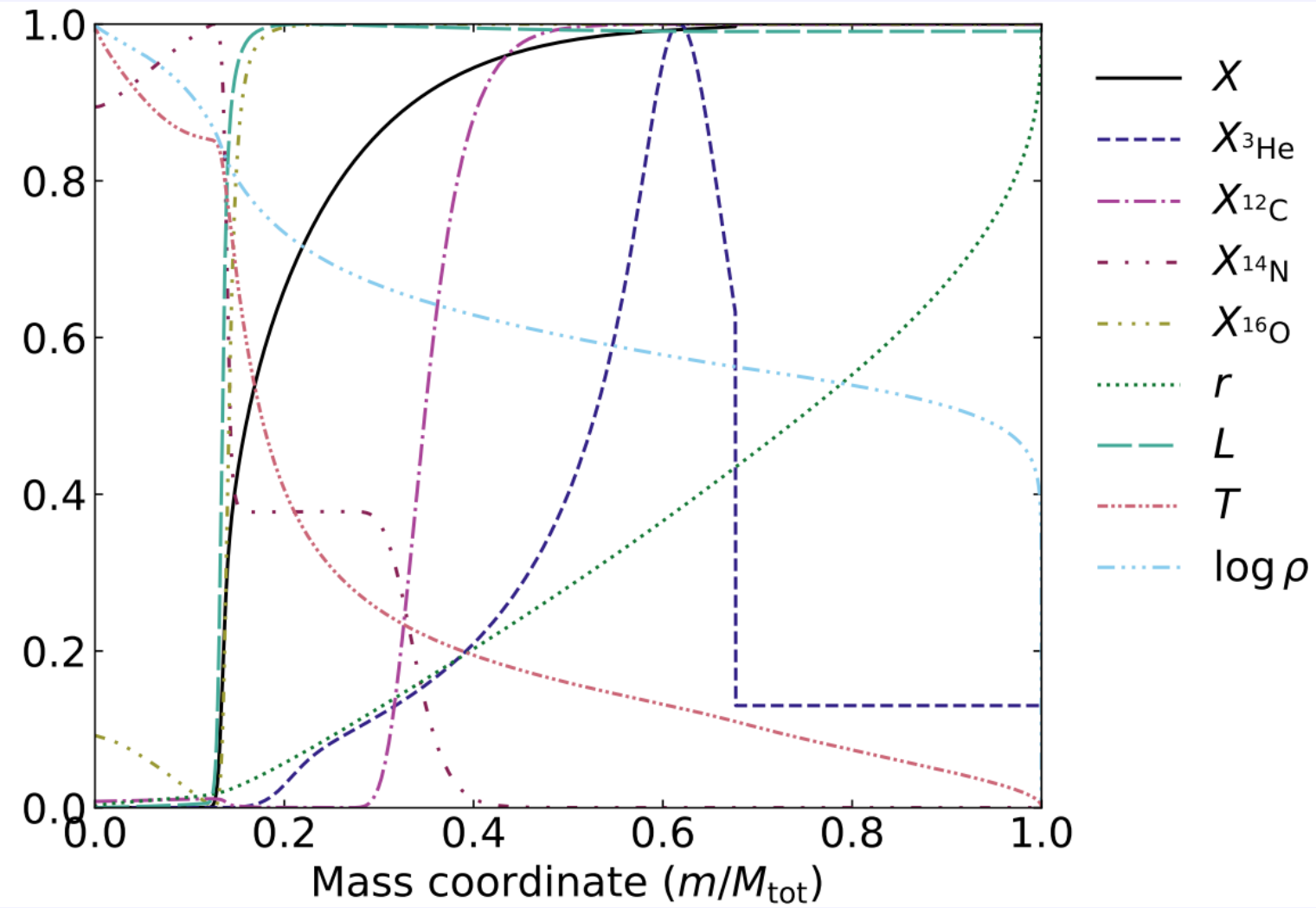
- He core grows gradually
- Initially, $q_c < q_{SC}$, so the core is stably growing on the nuclear timescale
- Remember lecture 17 - core density is higher after H burning for lower-mass stars. Core thus closer to being degenerate
- When $q_c > 0.1$, core contraction increases and the thickness of the shell rapidly decreases
- We're left with a very thin shell of H fusion

Stellar evolution - degenerate cores



Credit: JCD - Fig 12.7

Thin shell H burning (point 2)



Credit: JCD - Fig 12.13

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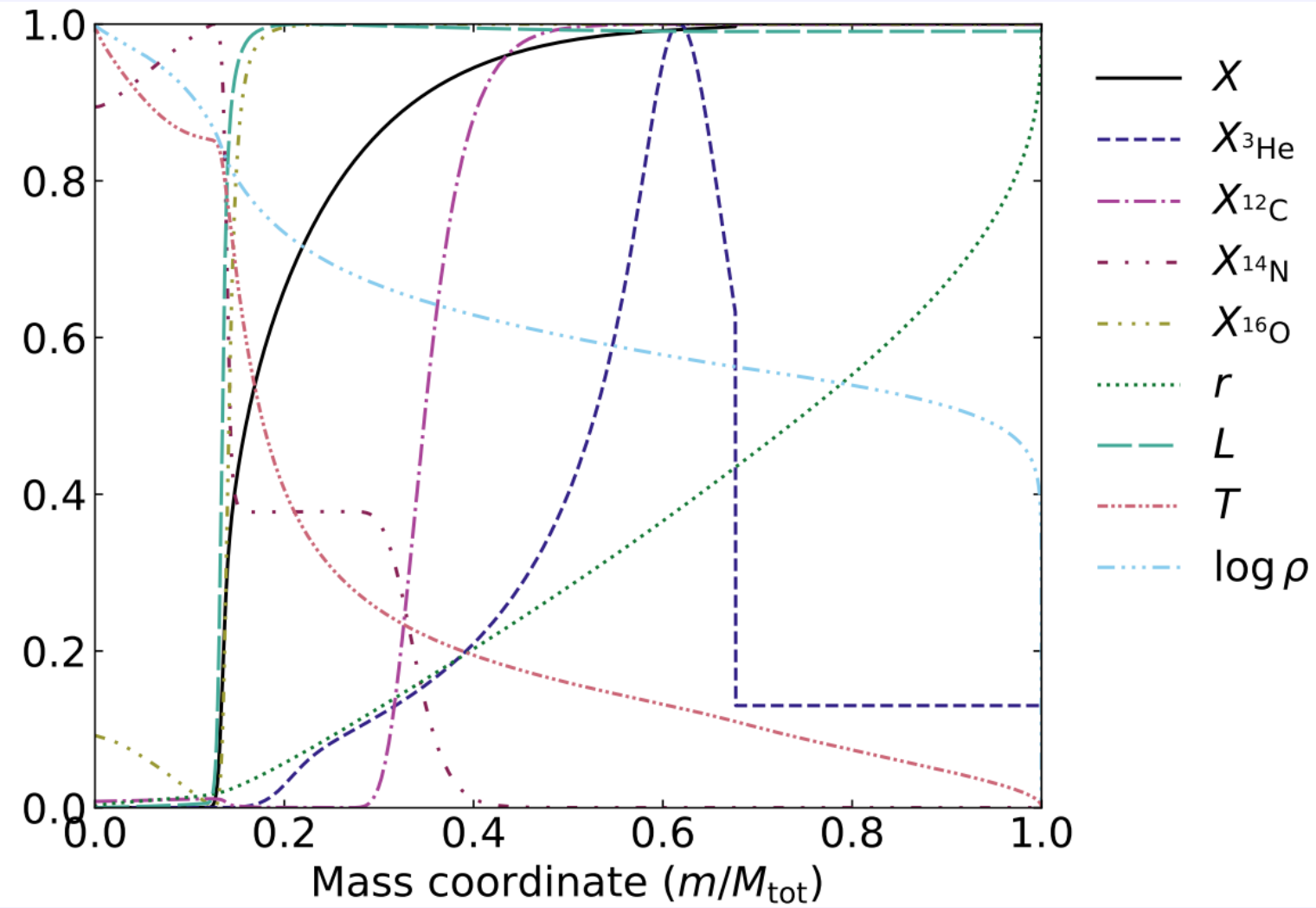
Thin shell H burning (point 2)

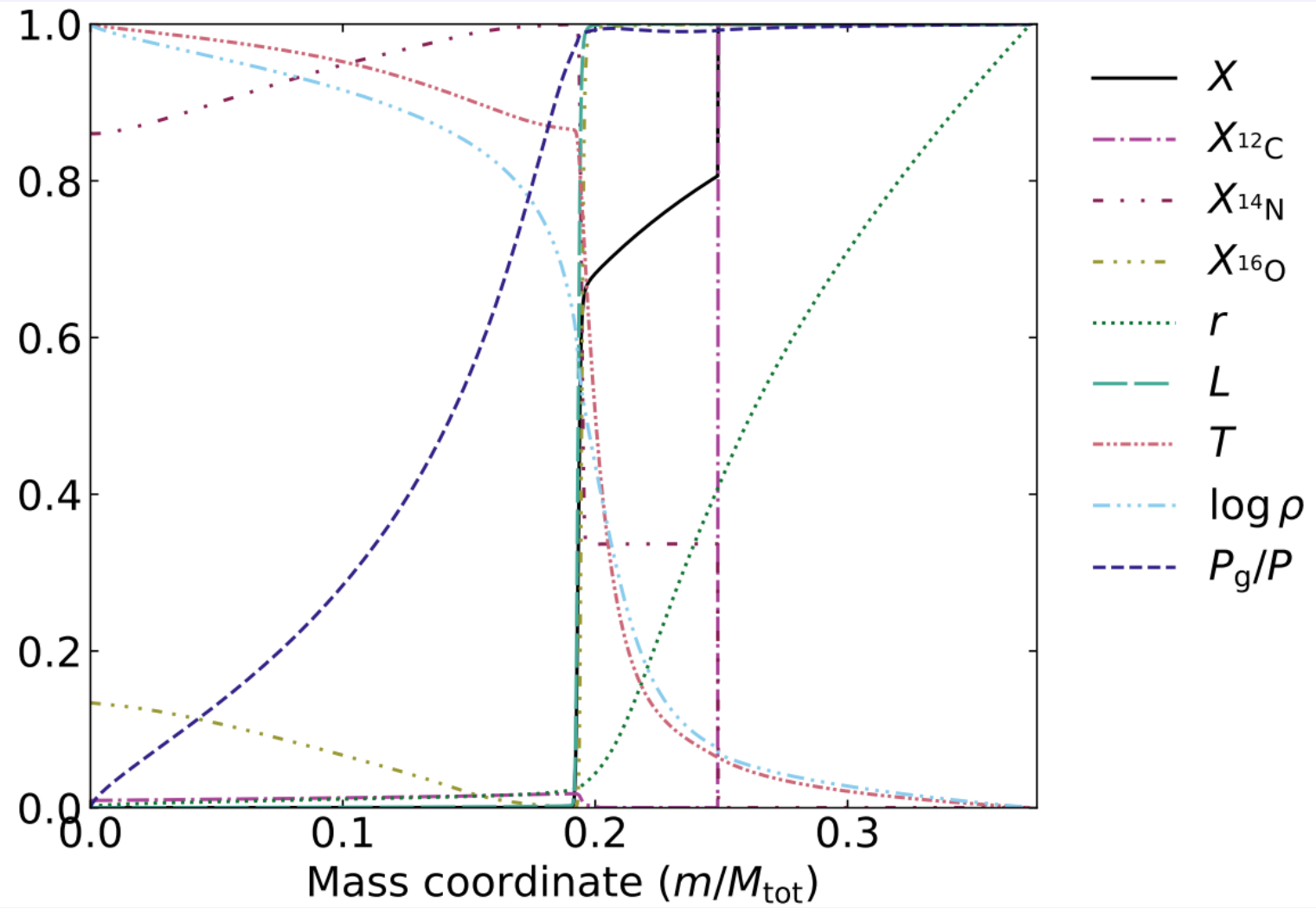


Credit: JCD - Fig 12.13

Core becomes degenerate

- Core contraction continues, which expands the star, which decreases the effective temperature
- This stops at the Hayashi tracks, after which the star moves upwards
- With density increasing even more in the core, the core becomes more and more degenerate
- Core nearly isothermal (energy in degenerate cores efficiently transported by conduction)
- As the star is on the Hayashi track, it's now developed a deep convective envelope
- This deep convective envelope changes the surface chemical composition

Degenerate core, deep convective envelope (point 3)



Credit: JCD - Fig 12.14

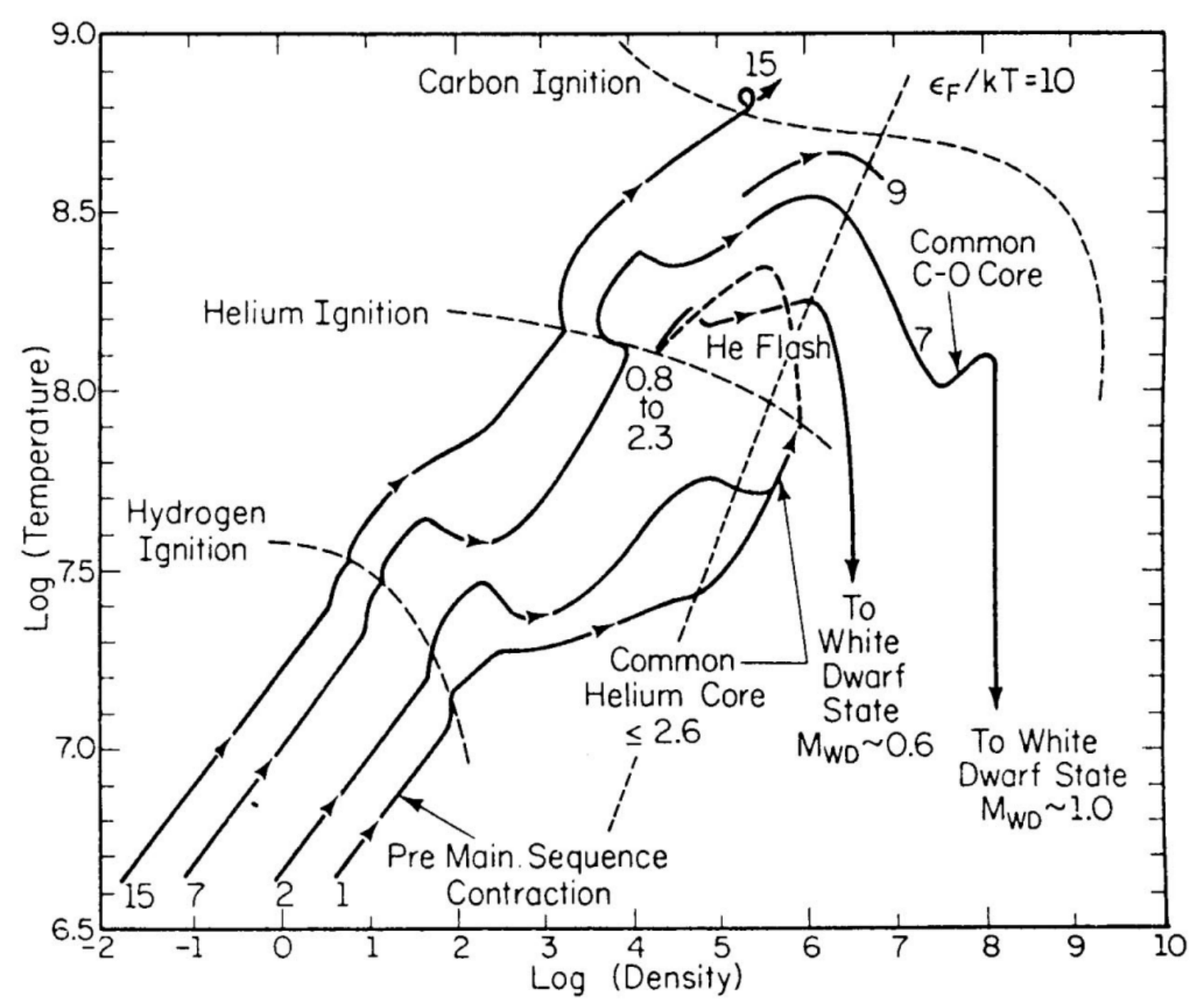
He flash

- Degenerate core means the temperature no longer increases from contraction [thermal energy of degenerate electrons is independent of temperature]
- However, the core temperature still increases due to the shell temperature increasing
- When the temperature reaches $\sim 10^8$ K, He fusion can start
- This happens in an unstable way on a thermal timescale, and is called the He flash

Reason for He flash

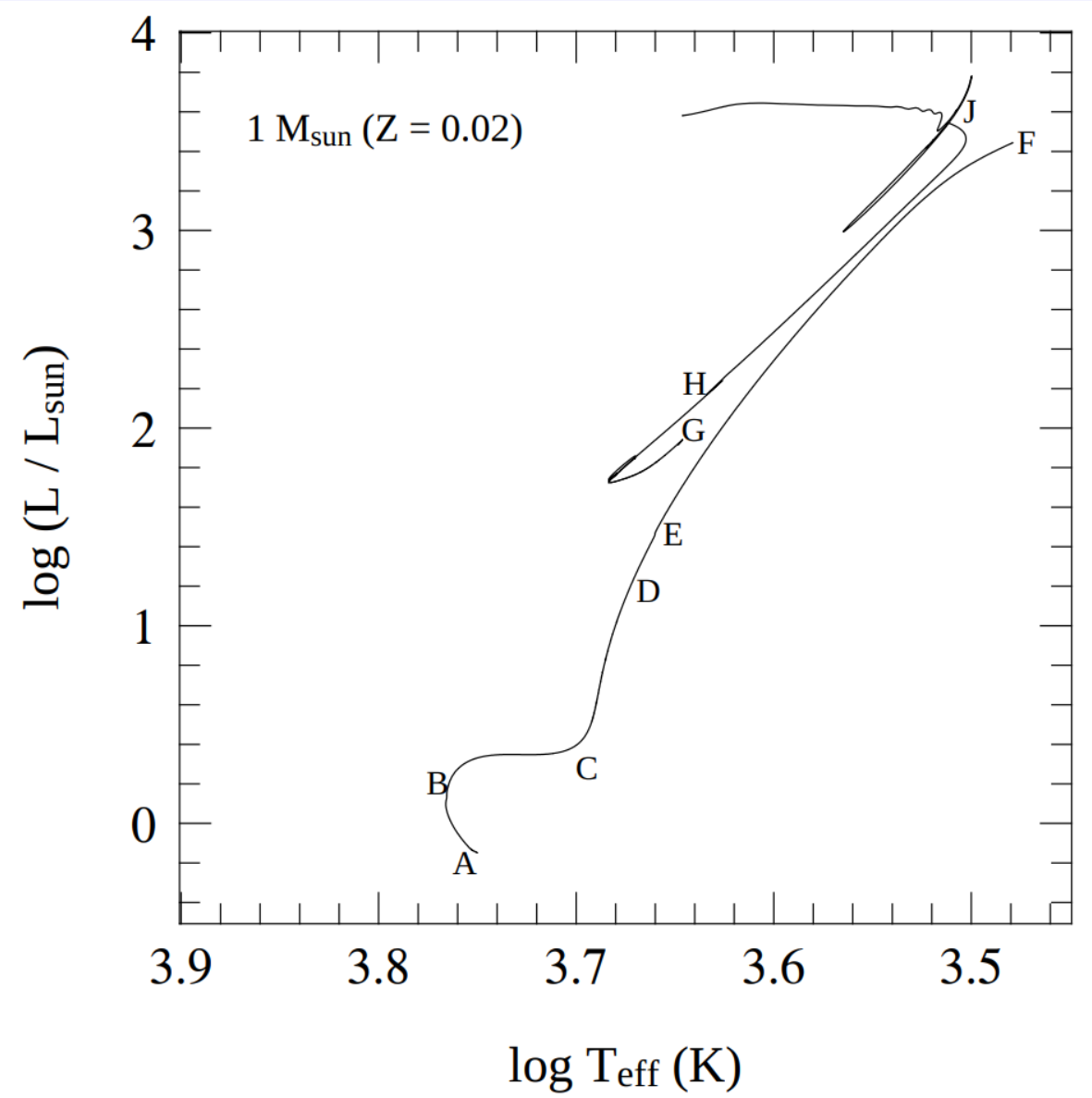
- In non-degenerate matter: release of energy from nuclear fusion leads to an increase in pressure and thus an expansion and cooling to remain in equilibrium
- In degenerate matter:
  - Pressure and temperature are no longer related
  - Thus nuclear fusion will increase temperature with no change to the core
  - This further speeds up energy generation and thus increases temperature even more
  - This leads to thermal runaway and only lasts a few hours
  - It stops when the temperature has reached such high temperatures that the degeneracy is lifted
  - The star now enters a phase of quiet He burning

Stellar evolution - degenerate cores



Credit: JCD - Fig 12.7

Low-mass stellar evolution - the He flash



Credit: Pols - Fig 10.5

Observing the He flash?

- Outer parts of the star can not react faster than the dynamical timescale, which is now on the order of months (large radius)
- The surface thus barely notices the violent internal behaviour of the short He flash
- However, once the He flash is over and the star has settled down, it has settled roughly on the same luminosity level for all low-mass stars
- This "He burning main sequence" is visible in the HR diagram as a Horizontal Branch.

Evolution after core He burning

- The process is now similar to intermediate-mass stars
- Core He is exhausted
- Two shells are established
- C-O core becomes degenerate
- Mass loss leads to planetary nebula
- Final result is a white dwarf

Summary

- Low-mass stars have no convective cores, so H exhausts gradually
- H shell first thick, then thin
- Core becomes degenerate prior to He fusion
- This leads to the He flash after which the star settles back down to equilibrium
- Subsequent evolution is the same as for moderate-mass stars

NEXT LECTURE: Stellar clusters and isochrones